

LAYER 4 · SYSTEM GOVERNANCE & ASSURANCE TIER [0.1–1 Hz]

Policy Supervision & Safety Cases

Ch 14 Shared Autonomy  
Policy Blending · Takeover

Ch 15 Verification  
Sim → PIL → HIL → In-Situ

Ch 16 Safety Cases  
Goal Structuring · UL 4600

Ch 17 Frontier Limits  
Observational Limits

LAYER 3 · THE BRAIN: COGNITIVE DELIBERATION TIER [1–50 Hz · Linux MPU / Edge NPU]

High-Capacity Foundation Models, Spatial Perception, Latent World Belief, and Action Chunking (Ch 03 Brain)

1. Ingestion (Ch 05)  
MIPI CSI-2 · Buffers

2. Perception (Ch 08)  
 $o_t \rightarrow z_t \cdot \text{ViT Encoders}$

3. Memory (Ch 09)  
 $z_t \rightarrow b_t \cdot \text{SE(3) Belief}$

4. Intent (Ch 10)  
 $b_t \rightarrow L_t \cdot \text{TTL Lease}$

5. Planner (Ch 11)  
 $L_t \rightarrow \hat{u} \cdot \text{ACT} / \text{Chunk}$

----- PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (Shared SRAM Mailbox · Untrusted  $\hat{u}_{t:t+H} \rightarrow \text{Verified } u^*$ ) -----

Proposal  $\hat{u}$

LAYER 2 · THE NERVOUS SYSTEM: REAL-TIME SAFETY & TIMING TIER [1000 Hz · Bare-Metal MCU]

Zero-Allocation Deterministic Silicon, Hardware Watchdogs, and Safety Refusal (Ch 04 Nervous System)

1 kHz Reflex Enforcer (Ch 12)  
Control Barrier Filter  $h(\mathbf{x}) \geq 0 \cdot \text{Veto } \hat{u}$

Silicon Placement & Bus QoS (Ch 13)  
Interconnect QoS · SRAM Seqlock · Timers

Certified Actuator Latch  
Latches  $u^*$  into PWM Compare Registers

LAYER 1 · THE PHYSICAL BODY & CONTINUOUS PLANT [Continuous Dynamical Mechanics]

Matter, Inertia, Transduction, Friction, and Thermal Limits (Ch 01 Boundary · Ch 02 Body · Ch 06/07 Reality Gap)

Power Inverters & Gate Drivers  
PWM Switching → Phase Current  $i_{\text{phase}}$

Mechanics, Torque & Momentum  
Lorentz Force → Torque  $\tau \rightarrow \text{Momentum } p = mv$

Causal Boundary & Dissipation  
Stopping Envelope  $d_{\text{stop}} \leq d_{\text{clear}} \cdot \text{Heat}$

Endogenous Sensory Shift ( $o_{t+1}$ )